

Total No. of Questions : 4]

SEAT No. :

PB-405

[Total No. of Pages : 2

[6270]-224

B.E. (Honors) (Insem)

**APPLICATIONS DEVELOPMENT USING AUGMENTED
REALITY AND VIRTUAL REALITY**

(2019 Pattern) (Semester - VIII) (410703)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Give significance of Gamification and explain the categories of Gamification with example. [8]
- b) Explain the virtual reality? Explain components of VR technologies. [7]

OR

- Q2)** a) What is AR toolkit? Demonstrate Integration of AR toolkits with existing IDE's [8]
- i) Unity-Vuforia,
 - ii) Visual Studio, with suitable steps
- b) Explain the Augmented Reality? Explain components of AR technologies. [7]

- Q3)** a) Write short note on : [9]
- i) VR SDK'S
 - ii) Open VR SDK,
 - iii) Google VR SDK
- b) What are the Hardware and Software requirements for VR? Discuss how Controllers are used in VR. [6]

OR

P.T.O.

Q4) a) Write short note on :

[9]

- i) VR SDK'S
- ii) Stream VR SDK,
- iii) VRTK

b) What is Unity XR API's? Why it is required in any VR project?

[6]

